
Artificial Intelligence Engineer

A career talk for senior secondary students. The pathways, the work, and the future of building intelligent systems.

AUDIENCE

Classes IX to XII

VENUE

KV RK Puram, Sector 2

DEPARTMENT

DEPFE, NCERT, New Delhi

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B.Tech in Computer Science Engineering
Noida Institute of Engineering and Technology, 2020 to 2024

RoboGPT

Web applications that make robotic arm programming easier for non-technical users.

Robot-control interfaces

Real-time 3D visualisation and connected systems using gRPC and WebSockets.

From software to deployment

Python, TypeScript, React, Next.js and cloud infrastructure for real client projects.

What does Orangewood Labs do?

Robotic arms and software that make automation easier to use.

Robotic Arms

Hardware, control systems and developer tools

RoboGPT

Natural language interfaces for robot control

Automation Solutions

Robots applied to real manufacturing tasks

Robotic Arms

OWL 6.5 brings together the robotic arm, control systems and an SDK so developers can program and integrate the robot.

HARDWARE + CONTROLS + SOFTWARE



RoboGPT

Instead of programming every robot movement manually, use natural language or voice commands and AI-assisted workflows to teach or control the robot.

My work: building the web application and robot-control interfaces.

EXAMPLE WORKFLOW

"Pick up the red apple."

1. Describe the task in plain language.
2. Use the interface to review the intended action.
3. Run under appropriate safety checks and supervision.

Automation Solutions

Put robotic arms to work on repetitive tasks by combining hardware, software and application-specific integration.

Coating and painting

Powder coating and spray painting applications for repeatable surface finishing.

Visual inspection

Vision-based checks that support quality control and help identify defects.

Bin picking

Picking and moving parts from a bin to support material handling.

What does a career in AI really involve?

A structured walk through everything you need to decide whether this profession is right for you.

01 Educational Pathway

Two routes: degree or self taught

02 Nature of Work

Day to day tasks and tools

03 Working Conditions and Attributes

Environment and skills that help

04 Career Progression and Salary

Growth from entry to leadership

05 Opportunities and Colleges

Where to work and where to study

06 Courses and Future Outlook

Building employability for the long run

Getting into the field

There is no single fixed route. Whether through a degree or by building on your own, a strong grounding in mathematics and programming matters more than any one credential.

01

EDUCATIONAL PATHWAY

Two ways into the field

There is no single fixed route. Pick the one that fits you, or blend both.

ROUTE A

The degree route

Structured, credential first

- 1 Science with Maths in Class XI and XII**
Add Computer Science if offered, and start Python early.
- 2 Crack an entrance exam**
JEE, BITSAT or CUET for IITs, NITs, IIITs and BITS.
- 3 B.Tech or B.Sc in CSE, AI-ML or Data Science**
Optional M.Tech or PhD for research and senior roles.
- 4 Internships, then a full time job**
Campus placements and referrals open the first role.

ROUTE B

The builder route

Self taught, portfolio first

- 1 Learn one language deeply**
Master Python and the maths behind machine learning.
- 2 Build real projects**
Ship them on GitHub and let the work speak for you.
- 3 Find clients and freelance**
Take paid gigs, small contracts and open source work.
- 4 Improve, specialise, grow**
Turn a real track record into full time and senior roles.

What an AI engineer actually does

Core Responsibilities

- Design, train and fine tune **machine learning and deep learning** models
- Collect, clean and prepare data, then engineer features
- Deploy models into products and **monitor them in production**
- Work on NLP, computer vision, recommendations and large language models
- Write reliable, production grade code and run experiments

Tools and Technologies

- **Languages.** Python, SQL, sometimes C++ or Java
- **Frameworks.** PyTorch, TensorFlow, scikit learn, Hugging Face
- **Cloud and DevOps.** AWS, GCP, Azure, Docker, Git
- **Data.** Pandas, NumPy, Spark, vector databases
- Collaborates with data scientists, product and software teams

The environment and what helps you succeed

Hours and Setup

- Typical **40 to 45 hour** work weeks
- Extra hours near launches or deadlines
- **Hybrid and remote** work are common
- On site work for robotics and hardware

Teams and Culture

- Small **cross functional teams**
- Agile sprints, code reviews, stand ups
- Global collaboration across time zones
- Constant need to keep learning

Personal Attributes

- Strong **mathematics** and logical reasoning
- **Curiosity** and self driven learning
- Patience for experimentation and debugging
- Explaining complex ideas **simply**

Growth and reward

The path runs from intern to leadership. Compensation rises quickly with proven delivery and technical ownership.

02

PROGRESSION AND SALARY

From entry level to leadership

Intern or Trainee

Starting out

Learn on real projects, assist with data and experiments while completing studies.

Junior AI or ML Engineer

0 to 2 years

Build and test models under guidance. Strengthen coding and data skills.

AI or ML Engineer

2 to 5 years

Own features end to end, deploy models to production, mentor juniors.

Senior AI Engineer

5 to 8 years

Lead complex systems, set technical direction, review architecture.

Lead, Principal or ML Architect

8 to 12 years

Design large scale AI systems and guide multiple teams.

Head of AI, Director or CTO

12 plus years

Own AI strategy and organisation wide impact.

Indicative compensation in India

Entry or Junior 0 to 2 years	First roles. Varies widely by employer and city.	6 to 12 LPA
Mid level 2 to 5 years	Proven delivery and production experience.	12 to 25 LPA
Senior 5 to 8 years	Technical ownership and mentoring.	25 to 45 LPA
Lead or Principal 8 to 12 years	Architecture and multi team leadership.	45 to 80 LPA
Director or Head of AI 12 plus years	Strategy. Top product firms pay well above.	80 LPA plus

Figures are broad, indicative annual ranges in Indian Rupees and vary significantly with company, location, skills and global roles. Top product companies and overseas positions can pay considerably more.

Where the demand is

Sectors

- IT and software, e commerce
- Fintech, healthcare, automotive
- Defence, agriculture, telecom
- Edtech, media and entertainment

Organisations

- Global. Google, Microsoft, Amazon, NVIDIA, Meta
- Indian IT. TCS, Infosys, Wipro
- Startups. Sarvam AI, Krutrim and more
- Research. ISRO, DRDO, IITs, IISc

Geographies

- Bengaluru, Hyderabad, Pune
- Gurugram, NCR, Chennai
- Global. US, Europe, Singapore, UAE
- Remote, freelance and consulting

Recommended institutions

Premier Engineering Institutes

- **IITs.** Bombay, Delhi, Madras, Kanpur, Kharagpur, Hyderabad, which has a dedicated AI department
- **IISc Bangalore** for leading AI research
- **NITs.** Trichy, Surathkal, Warangal
- **BITS Pilani.** Pilani, Goa, Hyderabad

Strong CS and AI Programmes

- **IIITs.** Hyderabad, Delhi, Bangalore
- Delhi University, DTU and NSUT
- VIT, Manipal, Ashoka University
- Reputed state and private universities with AI or DS courses

Building employability for the long run

Courses and Certifications

- DeepLearning.AI specialisations by Andrew Ng
- Google, AWS and Azure ML certifications
- Stanford CS231n for vision, CS224n for NLP
- **Kaggle** competitions and Kaggle Learn

Internships and Portfolio

- Research internships at IITs and IISc
- Google Summer of Code and industry internships
- Open source and a **GitHub project portfolio**
- Build things and share them publicly

Future Outlook

- **Generative AI and agents** are reshaping every industry
- Government push through the **IndiaAI Mission**
- Rising focus on **responsible and ethical AI**
- One of the fastest growing, highest demand careers

IN SUMMARY

Build the future, thoughtfully.

Start with strong mathematics and coding, choose a pathway that fits you, keep building projects, and never stop learning. AI rewards curiosity and persistence.

Thank you. Questions and discussion.